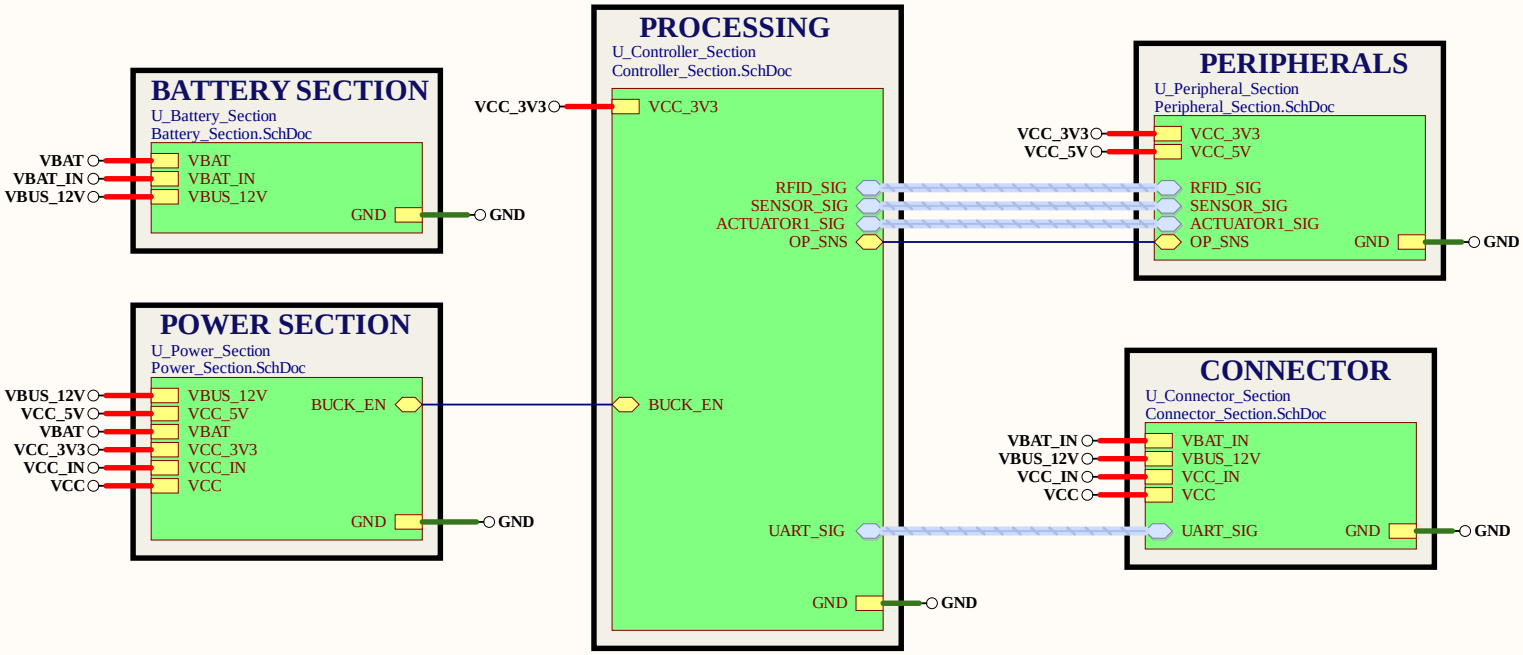
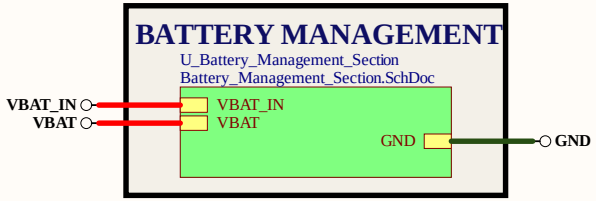
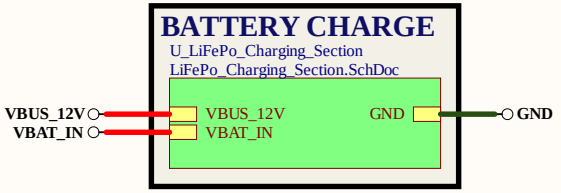
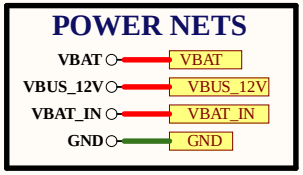


eSpark Consulting Group	Title: Francisco Gimenez.PrjPCB		
	Size: A4	Number: 1	Revision: 1
	Date: 3/22/2022	Sheet 1 of 13	
	File: E:\Git\..\Top_Sheet.SchDoc	Drawn By: Huzaifa Ali	

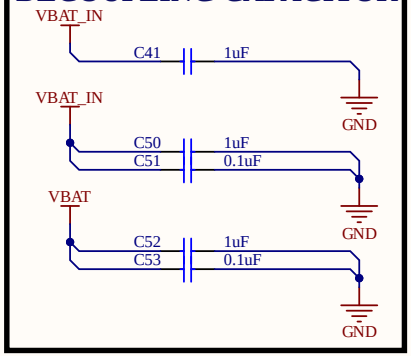




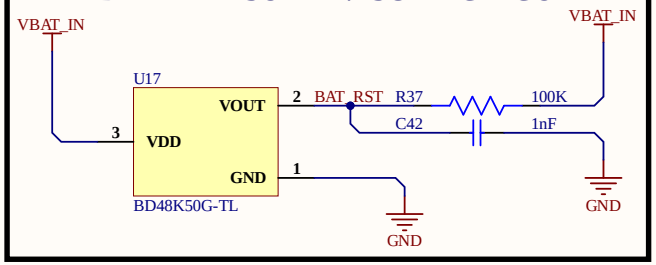


eSpark Consulting Group	Title: Francisco Gimenez.PrjPCB		
	Size: A4	Number: 1	Revision: 1
	Date: 3/22/2022	Sheet 2 of 13	
	File: E:\Git\..\Battery_Section.SchDoc	Drawn By: Huzaifa Ali	

DECOUPLING CAPACITOR

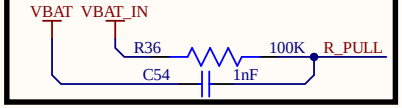


BATTERY SUPERVISORY CIRCUIT

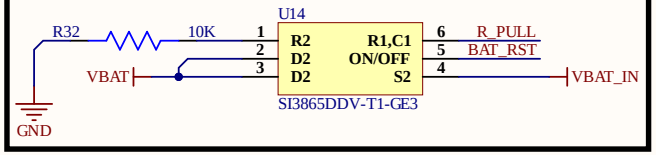


Detection voltage: 2.3V to 6.0V (Typ.)
Voltage threshold: 5V

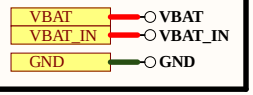
PULL UP RESISTOR



POWER SWITCH IC

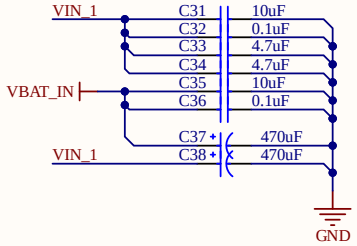


POWER NETS

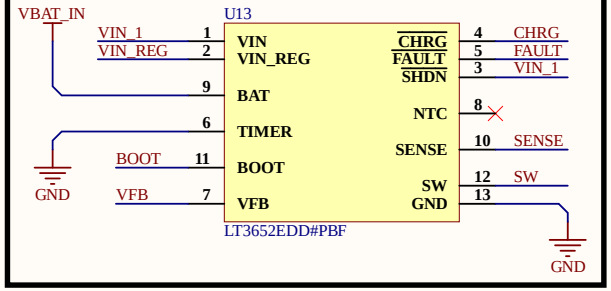


Title: Francisco Gimenez.PrjPCB			
Size: A4	Number: 1		Revision: 1
Date: 3/22/2022		Sheet 3 of 13	
File: E:\Git\..\Battery_Management_Section.SchDoc		Drawn By: Huzaifa Ali	

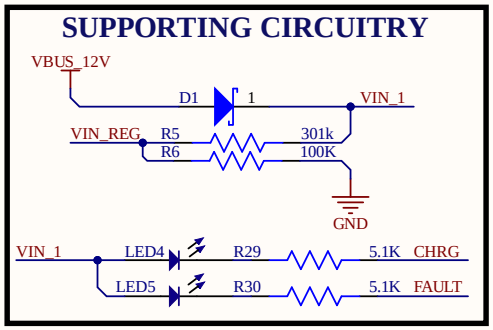
DECOUPLING CAPACITORS



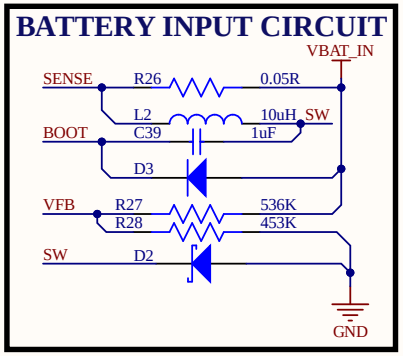
BATTERY CHARGING CIRCUIT



SUPPORTING CIRCUITRY

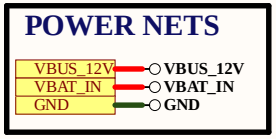


BATTERY INPUT CIRCUIT

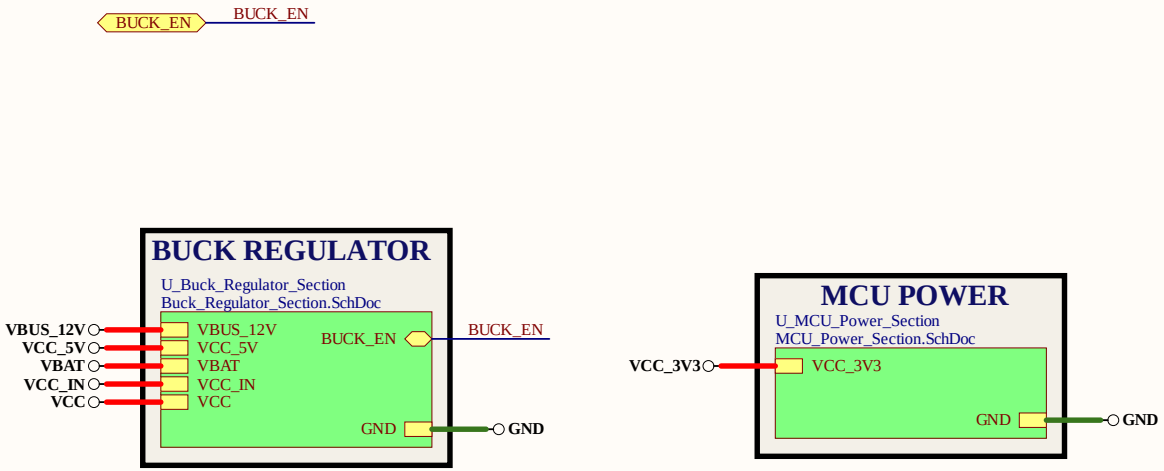


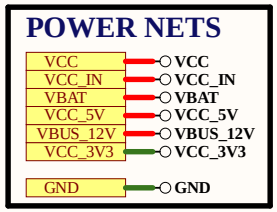
Battery Chemistry: 2 Cell LiFeOP4
Input Voltage: 12V USB
Battery Voltage: (2 × 3.6V) = 7.2V
Charge current: 2A

POWER NETS



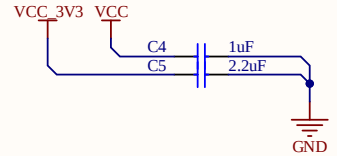
eSpark Consulting Group	Title: Francisco Gimenez.PrjPCB		
	Size: A4	Number: 1	Revision: 1
	Date: 3/22/2022	Sheet 4 of 13	
	File: E:\Git\..\LiFePo_Charging_Section.SchDoc	Drawn By: Huzaifa Ali	



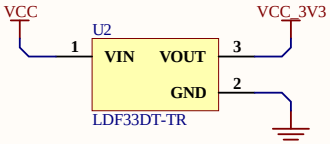


Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet 5 of	13
File:	E:\Git\..\Power_Section.SchDoc	Drawn By:	Huzaifa Ali

DECOUPLING CAPACITOR



3V3 REGULATOR



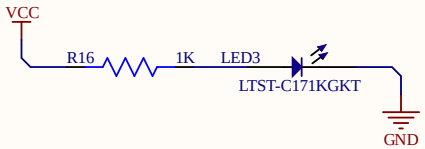
Input Voltage: 12V USB or 7.2V VBAT
Output Current: 1A
Output Voltage: Fixed 3.3V
Voltage Drop: 500 mV max. at 1 A load

- Calculator value for current limiting Resistance: 480R for 20mA @VCC=12V

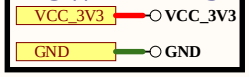
- Closest standard value=470R

LED max current rating=30mA (Datasheet =pg:3)
LED test current= 20mA (pg:5)

POWER INDICATION LED



POWER NETS

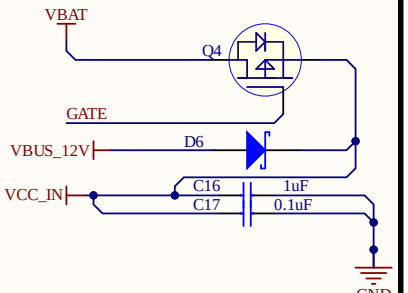


Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet6 of	13
File:	E:\Git\..\MCU_Power_Section.SchDoc	Drawn By:	Huzaifa Ali

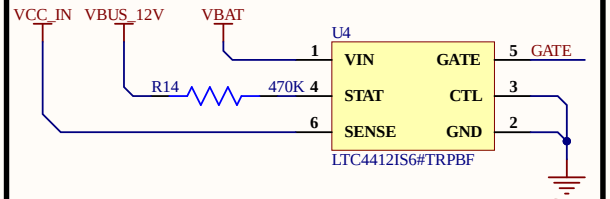
BUCK ENABLE



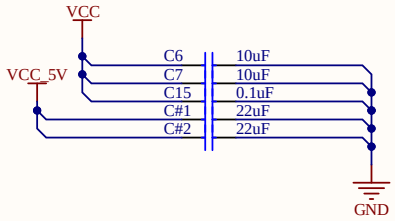
SUPPORTING CIRCUITRY



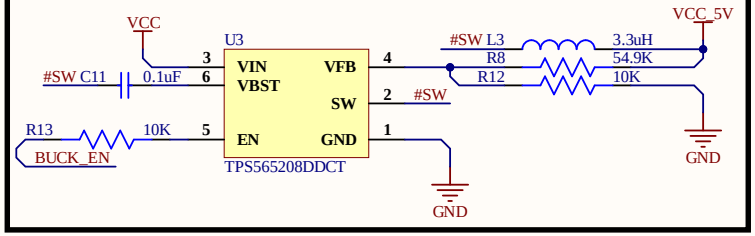
OR CONTROLLER SECTION



DECOUPLING CAPACITOR



BUCK SWITCHING REGULATOR CIRCUIT



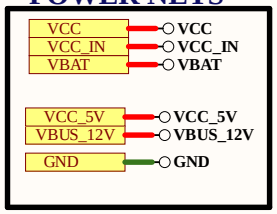
3.3uH inductor and output decoupling capacitor missing in the current design. (pg: 1,14.... datasheet)

- Input Voltage: switches between 12V USB and VBAT (7.2V)
- Output Voltage: 5V
- Output Current: 5A Maximum.

Output Voltage Calculation:

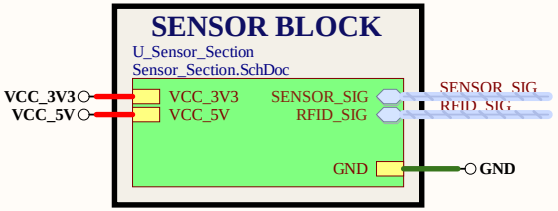
Formula: V =0.760 (1+R1/R2) (pg:13)
R1= 54.9K, R2=10K, L=3.3uH (pg:14)
VOUT= 4.93V
UVLO threshold: 3.6V (pg:5)

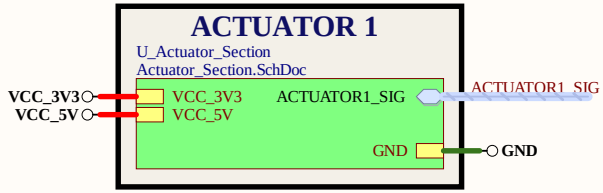
POWER NETS

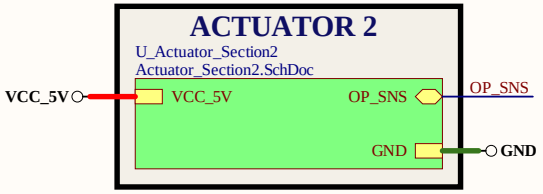


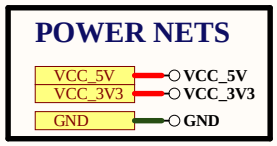
Title: Francisco Gimenez.PrjPCB			
Size: A4	Number: 1		Revision: 1
Date: 3/22/2022		Sheet 7 of 13	
File: E:\Git\..\Buck_Regulator_Section.SchDoc		Drawn By: Huzaifa Ali	







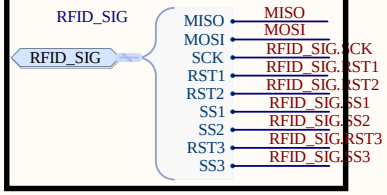




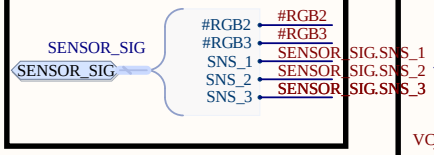
Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet8 of	13
File:	E:\Git\..\Peripheral_Section.SchDoc	Drawn By:	Huzaifa Ali

SNS signals are test signals for motor and RGB LED.

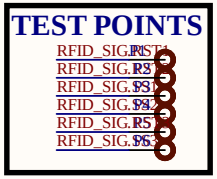
RFID INPUTS



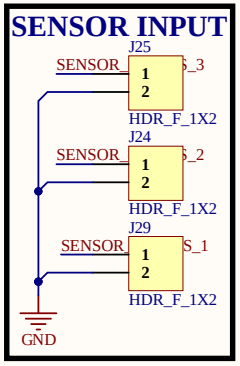
SENSOR SIGNAL



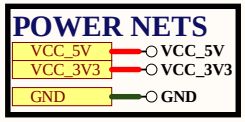
TEST POINTS



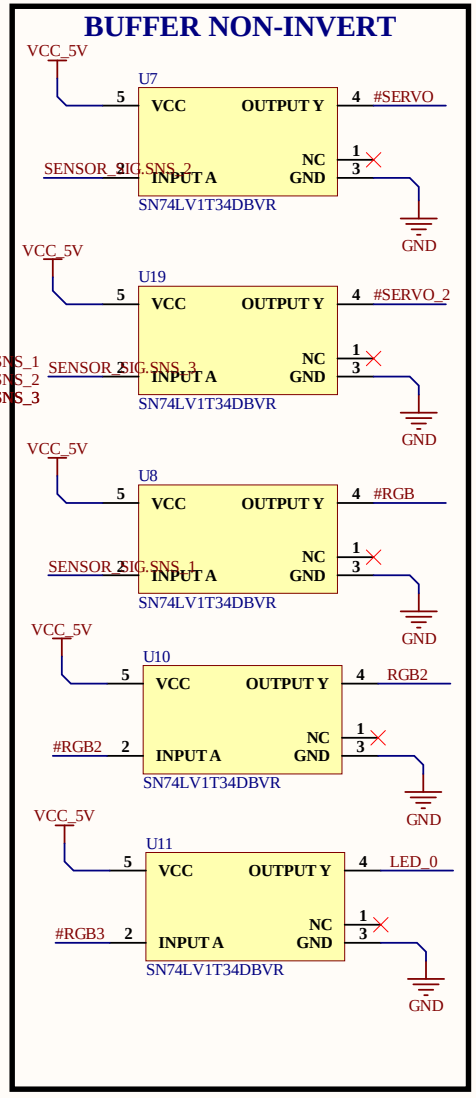
SENSOR INPUT



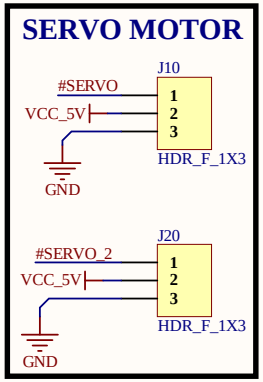
POWER NETS



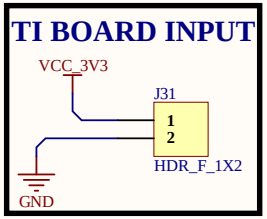
BUFFER NON-INVERT



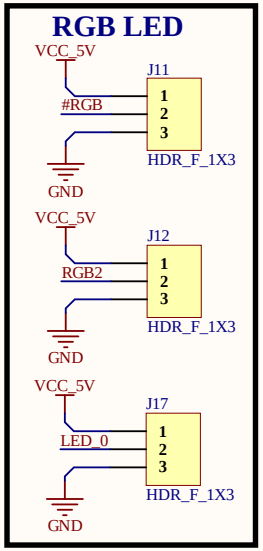
SERVO MOTOR



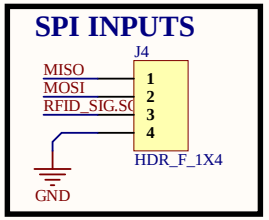
TI BOARD INPUT



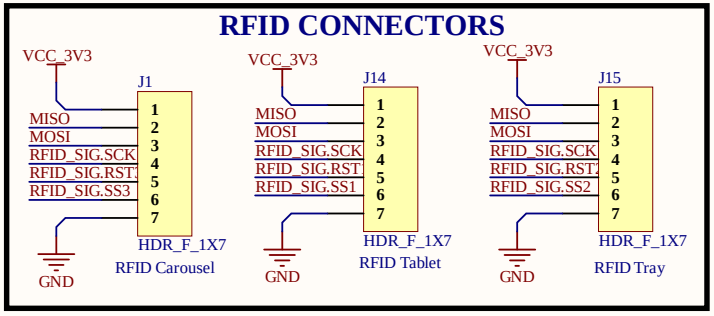
RGB LED



SPI INPUTS



RFID CONNECTORS

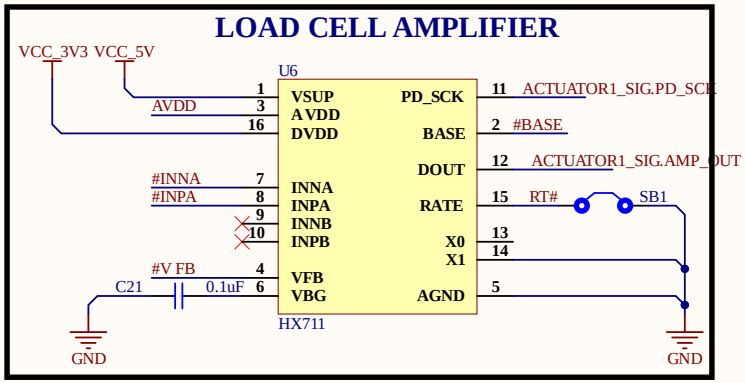


Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet 9 of	13
File:	E:\Git\..\Sensor_Section.SchDoc	Drawn By:	Huzaifa Ali

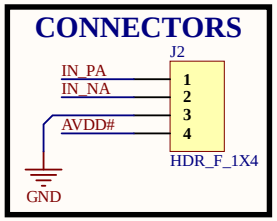
ACTUATOR1 SIGNAL



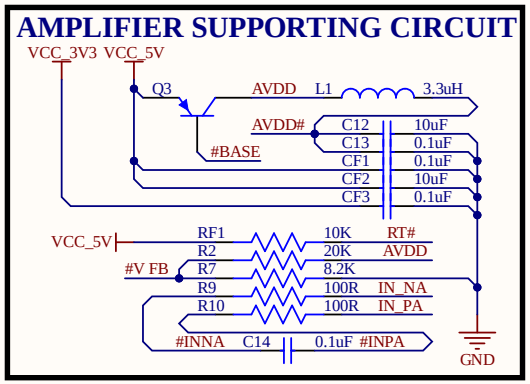
LOAD CELL AMPLIFIER



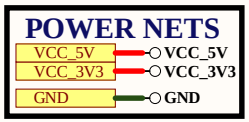
CONNECTORS



AMPLIFIER SUPPORTING CIRCUIT



POWER NETS

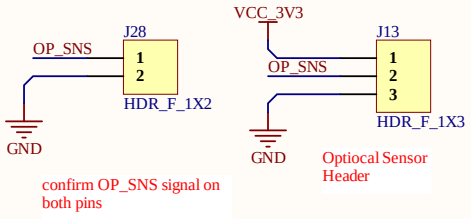


Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet 10 of	13
File:	E:\Git\..\Actuator_Section.SchDoc	Drawn By:	Huzaifa Ali

ACTUATOR2 SIGNAL

OP_SNS — OP_SNS

SENSOR INPUT CONNECTORS

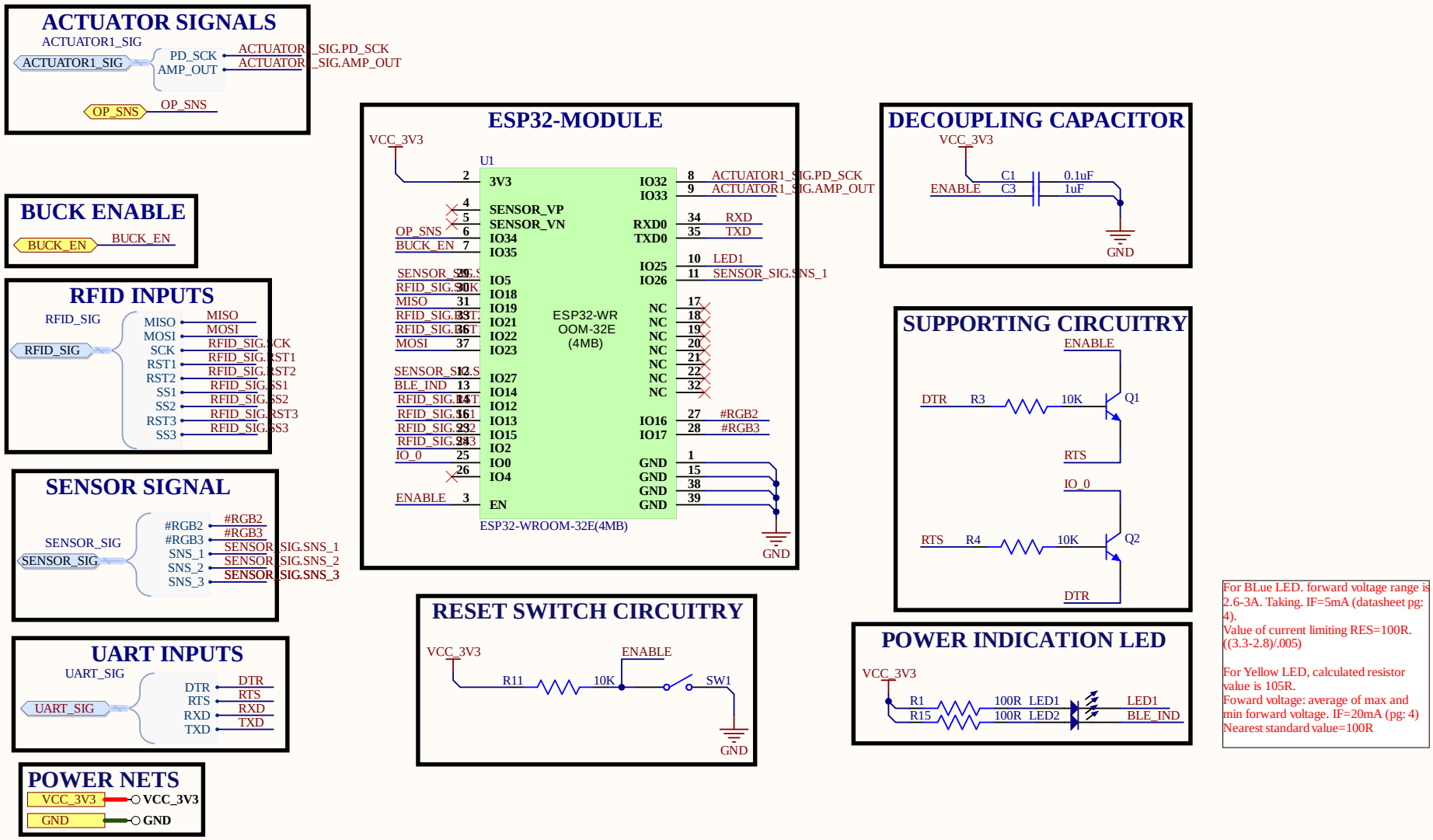


POWER NETS

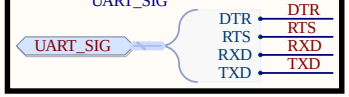
VCC_5V — VCC_5V

GND — GND

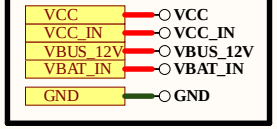
Title	Francisco Gimenez.PrjPCB		
Size	Number		Revision
A4	1		1
Date:	3/22/2022	Sheet 1 of	13
File:	E:\Git\..\Actuator_Section2.SchDoc	Drawn By:	Huzaifa Ali


ACTUATOR SIGNALS
ACTUATOR1_SIG
ACTUATOR1_SIG
PD_SCK
AMP_OUT
ACTUATOR1_SIG.PD_SCK
ACTUATOR1_SIG.AMP_OUT
OP_SNS
OP_SNS
BUCK ENABLE
BUCK_EN
BUCK_EN
RFID INPUTS
RFID_SIG
MISO
MOSI
SCK
RST1
RST2
SS1
SS2
RST3
SS3
SENSOR SIGNAL
SENSOR_SIG
SENSOR_SIG
#RGB2
#RGB3
SNS_1
SNS_2
SNS_3
SENSOR_SIG.SNS_1
SENSOR_SIG.SNS_2
SENSOR_SIG.SNS_3
UART INPUTS
UART_SIG
UART_SIG
DTR
RTS
RXD
TXD
POWER NETS
VCC_3V3
VCC_3V3
GND
GND
ESP32-MODULE
VCC_3V3
U1
3V3
SENSOR_VP
SENSOR_VN
IO34
IO35
IO5
IO18
IO19
IO21
IO22
IO23
IO27
IO14
IO12
IO13
IO15
IO2
IO0
IO4
EN
ESP32-WR
OOM-32E
(4MB)
IO32
IO33
RXD0
TXD0
IO25
IO26
NC
GND
ESP32-WROOM-32E(4MB)
GND
DECOUPLING CAPACITOR
VCC_3V3
ENABLE
C1
C3
0.1uF
1uF
GND
SUPPORTING CIRCUITRY
ENABLE
DTR
R3
10K
Q1
RTS
IO_0
RTS
R4
10K
Q2
DTR
RESET SWITCH CIRCUITRY
VCC_3V3
ENABLE
R11
10K
SW1
GND
POWER INDICATION LED
VCC_3V3
R1
R15
100R
LED1
100R
LED2
LED1
BLE_IND
For BLue LED. forward voltage range is 2.6-3A. Taking. IF=5mA (datasheet pg: 4).
Value of current limiting RES=100R. ((3.3-2.8)/.005)
For Yellow LED, calculated resistor value is 105R.
Foward voltage: average of max and min forward voltage. IF=20mA (pg: 4)
Nearest standard value=100R
eSpark Consulting Group
Title
Francisco Gimenez.PrjPCB
Size
A4
Number
1
Revision
1
Date:
3/22/2022
Sheet 12 of 13
File:
E:\Git\..\Controller_Section.SchDoc
Drawn By:
Huzaifa Ali

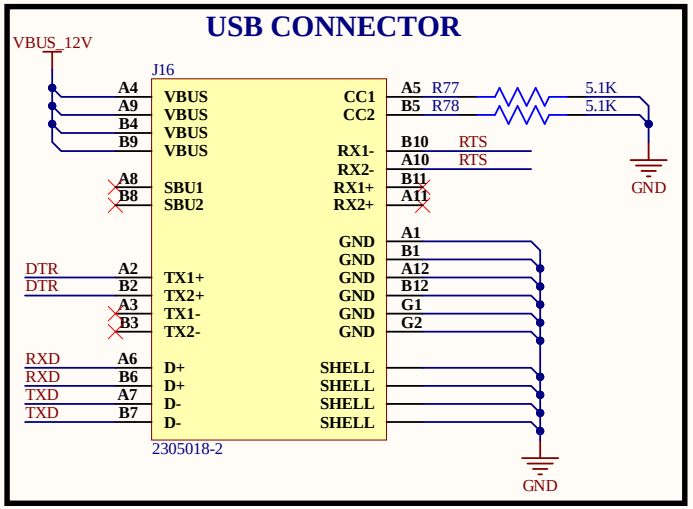
UART INPUTS



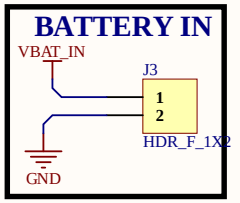
POWER NETS



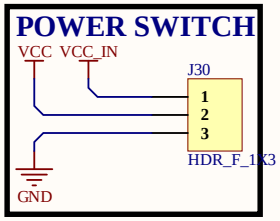
USB CONNECTOR



BATTERY IN



POWER SWITCH



VCC and VCC_IN are externally short circuited via a jumper on the board. VCC_IN is the main power input from source selector IC.

Title: Francisco Gimenez.PrjPCB			
Size: A4	Number: 1		Revision: 1
Date: 3/22/2022		Sheet 13 of 13	
File: E:\Git\..\Connector_Section.SchDoc		Drawn By: Huzaifa Ali	

Mounting Hole Size (Dia): 3.1mm
Estimated Maximum Height of PCB: 10.544mm
Number of Vias: 184

